Rowan McDaid

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Education

Vrije Universiteit *Sep 2025 – Expected Jul 2028*
BSc Computer Science (2nd year) Amsterdam, Netherlands

- Relevant modules: Object-Oriented and Functional Programming (in progress), Algorithms & Data Structures, Computer Programming, Computer Networks, Web Technology, Systems Programming Project, Requirements Engineering

Projects

YouTube Comment Ranker — github.com/slimjim46/Youtube-Extension | Chrome Web Store Jul 2026

- Shipped a Chrome extension (live on the Chrome Web Store) that ranks YouTube comments by toxicity severity, using a content script to detect the active video and a popup UI to fetch and display AI-scored results.
- Designed a Cloudflare Worker backend to securely proxy the YouTube Data API and OpenAI Moderation API, keeping all API keys server-side and out of the client.
- Implemented per-IP rate limiting, request caching (KV-backed, keyed by content hash), and adaptive batch scoring with exponential backoff to handle upstream rate limits gracefully.
- Stack: JavaScript, Chrome Extensions API (Manifest V3), Cloudflare Workers, KV storage, OpenAI Moderation API, YouTube Data API.

2D Lighting System — github.com/slimjim46/Retro-Game.git Jan 2025

- Built a tile-based sandbox game to explore 2D pixelated lighting, implementing a raycast light propagation system that calculates per-tile illumination from multiple light sources.
- Engineered shadow occlusion by ray-marching from each light source to every tile in radius, blocking light along paths that intersect solid blocks; rendered results via an HTML5 Canvas overlay using the `lighter` composite operation.
- Stack: HTML5 Canvas, JavaScript.

Pathfinding Snake — github.com/slimjim46/Pathfinding-Snake-Java Mar 2026

- Built a Snake game in Java (Swing) with a timer-driven game loop, collision detection, and game-over/restart handling.
- Implemented a breadth-first search (BFS) autopilot on a 30x30 grid that finds the shortest path to the food while avoiding walls and the snake's body, with a fallback move when no path exists.

Technical Skills

Languages	Java, Python, C, C++, JavaScript, SQL, Bash, Scala
Frameworks	React, Node.js
Tools	Git, MongoDB, VS Code
Concepts	Object-Oriented Programming, Algorithms & Complexity, TCP/IP, Client-Server Architecture, Memory Management, REST API